

EXPERIENCE

Flywl - Cloud Marketplace Startup

California, USA

AI Engineer

June 2025 - Present

- Engineered **Agentic RAG** pipeline using **S3 Vectors**, **OpenAI** indexing **35K+** documents, increasing decision time by **90%**
- Automated data normalization and preprocessing pipeline, de-duplicating **6K+** products across AWS, GCP, and Azure
- Designed robust system assigning unique IDs to **77K+** marketplace products, ensuring cross-platform consistency
- Built **full-stack** web app (**React, Flask**) hosting product catalog with fuzzy search, boosting matching accuracy by **20%**
- Deployed** the platform on **AWS** (EC2 + S3 + S3 Vectors), PostgreSQL to store activity data for **100+** internal users

Wisconsin Institute for Discovery

Wisconsin, USA

AI Research Engineer [\[Link\]](#)

January 2024 - June 2025

- Developed novel data selection strategies (task wise Uncertainty, Clustering) to identify most informative data points
- Fine tuned **LLM**- LLaMA-2 on **100K+** samples spanning **1800+** NLP tasks leveraging **CUDA GPUs** (HPC)
- Achieved significant reduction of **80%** in training data requirement, cutting down on annotation and training costs
- Evaluated performance using AlpacaEval (LLM-based evaluation tool), outperforming baseline in **53%** of test cases
- Validated task-wise selection via Hypothesis Testing, achieved **+4%** improvement in **accuracy** on benchmark datasets

Asurion/Synechron Technologies

Bangalore, India

AI/ML Engineer

January 2023 - June 2023

- Fine-tuned **LLMs** on **30K+** customer support chats to build Gen-AI powered troubleshooting assistant for tech queries
- Streamlined preprocessing pipeline for transcripts, reducing formatting time by **33%** through vectorized and regex ops
- Analysed **500K+** mobile insurance records using **SQL**, **Python** to uncover root causes behind declining conversion rates
- Collaborated with **cross-functional teams** to deliver **data-driven actionable insights** on peak-hour sales losses
- Created interactive **Tableau dashboards** for sales and marketing teams, increasing data-driven decision-making by **25%**
- Designed agent allocation rebalancing strategy, forecasting **~\$1.5M** sales improvement via simulation testing

Nanyang Technological University (NTU), Singapore

Singapore

AI/ML Researcher [\[Github\]](#)

August 2022 - December 2022

- Developed a robust end-to-end Neural Speaker Diarization (EEND) system leveraging **Pytorch** and SpeechBrain
- Integrated **CNN**-based feature encoder, and fine-tuned WavLM **Transformer** encoder, on speech dataset (**20+** hours)
- Achieved **5%** reduction in the Diarization Error Rate (DER) for above real life dataset

PROJECTS

Text-to-SQL Multi-Agent System with LangGraph

UW Madison (2025)

- Designed a multi-agent Text-to-SQL system in LangGraph with planner, schema retriever, SQL generator and executor
- Implemented efficient workflow across **3+** tables, enabling complex schema reasoning and reliable query generation
- Optimized agent feedback loops with validation and correction steps, reducing query failures, improving reliability

Big Data Systems [\[Github\]](#)

UW Madison (2025)

- Executed advanced **SQL** queries on Google **BigQuery** and trained linear regression model (**R²=0.87**) for loan forecasting
- Engineered **Spark-Hive** ETL pipeline and trained Decision Tree classifier with **89.5%** accuracy on loan approvals
- Built **ELT** pipeline with Airbyte, **Snowflake**, and **DBT** to integrate survey, stock, and forex data for unified data analysis.

PUBLICATIONS

- EMNLP MAIN 2025** - "Improving Task Diversity in Label Efficient Supervised Finetuning of LLMs." [arXiv:2507.21482](https://arxiv.org/abs/2507.21482)
Abhinav Arabelly*, Jagrut Nemade*, Jifan Zhang, Robert D Nowak

SKILLS

Programming & ML: Python, SQL, R, C++, PyTorch, TensorFlow, Transformers, CNN, XGBoost, LightGBM, Pandas, Numpy, Tableau, Matplotlib, Regression, scikit-learn

LLMs & NLP: LangChain, LangGraph, LLMs, RAG, Agentic Frameworks, FAISS, ChromaDB, Hugging Face, SpeechBrain

Cloud & MLOps: AWS, GCP, Spark, Kafka, Hadoop, BigQuery, Databricks, Docker, MLflow, FastAPI, Airbyte, Snowflake, DBT

EDUCATION

University of Wisconsin - Madison

Madison, WI

Master of Science in **Data Science**

August 2023 - May 2025

Courses: Artificial Intelligence, Machine Learning, Data Structures, Algorithms, Database Management, Big Data, Statistics

Birla Institute of Technology and Science (BITS), Pilani

Hyderabad, India

Integrated MSc. **Mathematics** and B.E. Electronics and Communication

August 2018 - May 2022